

Nonvalidation of mesospheric horizontal wind fluctuations derived from temperature data via comparison with MF radar wind measurements

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Abstract. Mesospheric transport processes and energy budget are tied closely to gravity wave activity. Any means of monitoring dynamics reliably can only contribute to our understanding of the physics of the mesosphere. In this study we employ temperature data derived from hydroxyl (OH) emission from around 85 km and combine it with the theory of *Hines* [1960] to investigate the practicality of deducing fluctuations in horizontal wind speed. *Myrabø and Harang* [1988] have used a simplified theory [*Hines*, 1965] but, apparently, to deduce tidal fluctuations in the horizontal wind; this prompts us to investigate the range of timescales for which the *Hines* [1965] result and the assumptions implicit therein may be used in practice. Our conclusion is that this theory may not be extended to scales of dynamics outside those intended by *Hines*. Thus while the *Myrabø and Harang* [1988] results seem plausible, they are probably incorrect.

1. Introduction and Underlying Theory

Gravity waves propagating upward enjoy an exponential increase in amplitude with height z according to $\exp(z/2H)$ (H being the scale height) in order to conserve energy. At some height, however, either convective or dynamic instabilities or both limit this exponential growth by removing sufficient energy from the wave field to restore stability. The waves continue to propagate upward in a saturated form until further limited. The resulting perturbations of the atmosphere may be described by equations (6)–(8) of *Hines* [1960] subject to the assumptions he specifies, i.e., adiabatic processes combined with small perturbations. Solutions of the *Hines* equations may be found in complex Fourier form whence the relation [*Hines*, 1960, equation (22)]:

$$\frac{p - p_0}{p_0 P} = \frac{\rho - \rho_0}{\rho_0 R} = \frac{U_x}{X} = \frac{U_z}{Z} = A \exp\left(\frac{z\gamma g}{2C^2}\right) \exp i(\omega t - k_x x - k_z z), \quad (1)$$

can be derived. Here P , R , X , Z , and A are known as polarization factors and are defined as follows [*Hines* 1960, equations (23)–(26)]:

$$P = \gamma \omega^2 \left[k_z - i \left(1 - \frac{\gamma}{2} \right) \frac{g}{C^2} \right],$$

$$R = \omega^2 k_z + i(\gamma - 1)gk_x^2 - i \frac{\gamma g \omega^2}{2C^2}, \quad (2)$$

$$X = \omega k_x C^2 \left[k_z - i \left(1 - \frac{\gamma}{2} \right) \frac{g}{C^2} \right],$$

$$Z = \omega(\omega^2 - k_x C^2),$$

where p_0 is the background pressure, p is the perturbed pressure, ρ_0 is the background density, ρ is the perturbed density, U_x is the horizontal velocity perturbation, U_z is the vertical velocity perturbation, ω is the circular frequency of the wave, g is the gravitational acceleration, k_x and k_z are the horizontal and vertical wave numbers, respectively, C is the speed of sound ($\approx 258 \text{ m s}^{-1}$), γ is the ratio of the specific heats ($= 1.4$), and i is the imaginary unit, here indicating any phase shift. Note that in the above and in the following equations we have used exactly the same notation as *Hines* [1960] to facilitate comparison, even though other conventions are more usual currently (e.g., lowercase c for speed of sound).

The wave numbers and the wave frequencies are related by the dispersion relation [*Hines*, 1960, equation (21)]:

$$\omega^4 - \omega^2 C^2 (k_x^2 - k_z^2) + (\gamma - 1)g^2 k_x^2 - \frac{\gamma^2 g^2 \omega^2}{4C^2} = 0. \quad (3)$$

It is examination of this dispersion relation that reveals the frequencies permitted to the waves: acoustic and internal gravity wave (GW) modes. In the Appendix we demonstrate how (1) may be combined with the ideal gas law to arrive at [*Hines*, 1965, equation (5)]

$$\frac{\Delta T}{T} = \frac{(P - R)}{X} U_x. \quad (4)$$

If we may assume that the vertical wavelength does not appreciably exceed the local scale height H , and where the wave

Table 1. Basic Parameters of Ebert-Fastie Spectrometer

Component of the Instrument	Description or Approximate Values
Geographic location	69.3°N, 20.3°E, geographic
Type of instrument	½ m Ebert-Fastie spectrometer
Focal length of the mirror, cm	50
Grating area, cm ²	≈100
Spectral order used	first order
Grating constant grooves, cm ⁻¹	12,000
Slit width, mm	≈0.25
Spectral resolution, nm	≈0.7
Slit height, cm	5
Field of view	≈6° × 6°
Detector type	RCA type with Ga As tube

period is substantially greater than the Brunt Väisälä period, it is possible to reduce this expression to [Hines, 1965, equation (6)]

$$\frac{\Delta T}{T} \approx \pm i \sqrt{\gamma - 1} C^{-1} U_x. \quad (5)$$

Again, this derivation is given via the Appendix; we feel it is important to fully understand the way the assumptions regarding k_z and wave period are incorporated into (5).

Wherever observing conditions have been conducive to obtaining time series of OH temperatures of more than 6 hours duration, tidal modes are clearly visible in the data. *Fritts and Vincent* [1987] discuss GW-tide interactions, and it is possible that such studies have prompted *Myrabø and Harang* [1988] to employ (5) to deduce horizontal velocity fluctuations corresponding to their observed OH temperature fluctuations. We note, however, that fluctuations corresponding to tidal timescales rather than GW timescales have been employed by *Myrabø and Harang* [1988]; while this satisfies the condition that the period should be significantly greater than the Brunt-Väisälä period, the vertical wavelength of semidiurnal and diurnal tides is expected to be considerably more than H (≈ 6.3 km typically). The review by *Forbes* [1995] describes vertical wavelength for tidal modes in detail: the (1, 1) mode exhibits a vertical wavelength around 4 times the scale height, semidiurnal modes have even longer wavelengths. Moreover, *Myrabø and Harang* [1988] consider timescales more akin to the semidiurnal tide. It is for shorter vertical wavelengths that the modification of eddy diffusion is most effective, and it is this that is related to the pseudotides [Vial and Forbes, 1989] with which *Myrabø and Harang* [1988] are concerned.

We shall now proceed to describe the OH measurements we use for a case study to investigate the range of timescales for which we are justified in using (5). We obtain wind information simultaneously from the University of Tromsø/University of Saskatchewan MF radar (described later), which we deem to be the actual neutral wind, and use it to validate (or refute) the OH-derived values.

2. OH Measurements

Since Meinel identified the OH band system in the night airglow in 1950, a large number of investigators have tried to elucidate the physics and chemistry behind the different patterns of behavior of the OH nightglow [Myrabø, 1984; Mulligan and Nallen, 1998; Choi et al., 1998]. The study of OH emissions alone or together with other nightglow emissions provides important information on temperature, density, composition, and

behavior of the upper mesosphere characteristic of the layer from which the emission originates. This layer is assumed to be centered on 85 km and ~ 5 km thick. OH emission measurements may also be used as an important tool for observing the effects of propagating gravity waves, through the atmosphere [Hatfield et al., 1981].

Recent developments in airglow imaging have demonstrated how intensities of emissions from the hydroxyl (OH) layer can yield such information on dynamics [e.g., Hecht et al., 1997; Hoppe, 1989]. Such imaging systems often observe only the airglow intensity, however, while use of a spectrometer enables us to derive temperatures [Sivjee et al., 1972; Suzaki and Tohmatsu, 1976]. The spectra of the night OH emission data employed in this work were part of the measurements undertaken during the months of January and February 1997 at a site close to Skibotn in northern Norway. Table 1 gives the description of the instrument (i.e., Ebert-Fastie spectrometer) we used for recording the data that we will consider in this study. At the time of our campaign we have been able to utilize limited periods of clear weather during which the instruments could obtain reliable for OH measurements, such as moonless conditions and little or no aurora.

The spectrometer was pointing toward zenith and the λ 835–855 nm region was scanned, and each scan covered the first-order wavelength regions in 50.31 s. In most cases an adequate signal-to-noise ratio was acquired by integrating 21 individual scans over 17.5 min.

Rotational temperature was calculated by Boltzmann distribution technique, using the intensity distribution of the P_1 (2), P_1 (3), P_1 (4), and P_1 (5) lines [Myrabø and Harang, 1988; Yizengaw, 1998]. Figure 1 shows an example of the time domain plot of rotational temperature derived from OH (6-2) Meinel band emission. In this figure we note two types of perturbation: we suggest these are the tidal perturbation, with larger amplitude (i.e., characterized by the whole period shown), and gravity wave perturbation, with smaller amplitude (most clear after midnight).

When investigating gravity waves using OH intensity information, Hoppe [1989] tells us that sensitivity is greatest when vertical wavelengths are greater than 10 km. At the same time, the theory presented above requires that vertical wavelengths should be less than this (one reviewer of this paper quotes $k_z \gg 1/2H$). Hatfield et al. [1981] also examine the effects of dynamics on the OH emission to essentially the same conclusion. When using temperature variation to investigate gravity

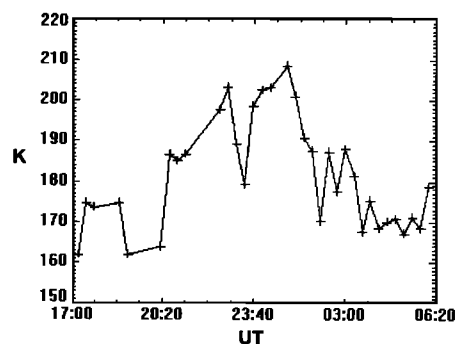


Figure 1. Time domain OH rotational temperature (K) on January 31 and February 1, 1997. This temperature is taken to be representative of an ~ 5 km layer centered on 85 km altitude.

Table 2. MF System Parameters

Parameter	Value
Geographic coordinates	69.58°N, 19.22°E
Operating frequency, MHz	2.8
Pulse repetition frequency, Hz	100
Peak transmitter power, kW	~20
Transmitter antenna beam width, deg.	17
Receiver antenna	three spaced inverted-V dipoles
Height resolution, km	3
Postintegration time, min	5

waves, apart from assuming the OH rotational temperature is representative of the neutral gas, we assume an adiabatic process and maintaining this condition is difficult for long periodicities and long wavelengths. We shall see, later on, the difficulties these almost mutually exclusive conditions make for applying (5) to real data.

3. MF Radar Wind Measurements

The joint University of Tromsø/University of Saskatchewan MF radar, using the “spaced antenna method,” is located in northern Norway at 69°N, 19°E. Table 2 gives the specification of the radar for the periods we will consider in this study, the most important points being the final height and time samplings of 3 km and 5 min, respectively. Winds are determined by the spaced antenna method elegantly described by *Hocking* [1997] using the full correlation analysis (FCA) described by *Briggs* [1984] and *Meek* [1980]. The radar operates unattended round the clock, so 288 profiles of signal fading time are available each day. Not each profile is complete, however, because of either poor signal-to-noise ratios (typical at night in winter) or due to total reflection of the signal (during periods of high auroral activity); indeed, we experience evidence for lack of signal in January and December. Our confidence in the FCA-derived winds being truly representative of the neutral dynamics is considerable not least because a wealth of data obtained by such systems over the years conforms well to theoretical expectations. *Hall et al.* [1998] most recently investigate spectral signatures in data from the MF system described here, addressing dynamics from tidal scales to the quasi-16-day wave. Very satisfactory comparisons between systems have been completed: *Manson et al.* [1992] for rockets, VHF, and European Incoherent Scatter (EISCAT) radars; *Manson et al.*

Table 3. OH Observation Campaign

Date		Duration, UT	
From	To	From	To
January 30, 1997	January 31, 1997	1600	0620
January 31, 1997	February 1, 1997	1700	0620
January 20, 1998	January 21, 1998	1600	0620

[1996] for a green line Fabry Perot Interferometer (FPI) (97 km); *Meek et al.* [1997] for an OH FPI (88 km). On the subject of tides alone, *Manson and Meek* [1991] represent one of many publications utilizing MF wind data. Comparisons with theory on dynamics at these scales may readily be made by examining, for example, *Forbes* [1995]. At shorter timescales, *Manson and Meek* [1997] use MF results to investigate gravity waves. Figure 2 shows an example of the time domain MF wind for the 3-km-wide range gate centered on 85 km altitude. Here also we observe perturbations similar to those seen in the rotational temperature. *Hocking* [1997] describes the difficulties in such wind measurements and, in particular, the element of uncertainty in height determination. One problem is that the refractive index is a function of electron density such that the radio wave gets progressively more retarded as it intrudes into the lower ionosphere. This effect, known as the group delay, means that heights of echoes are from lower down than apparent. *Hall* [1998] tried to correct for the effect. Fortunately, however, the correction is not likely to be more than a kilometer or so for the data in this paper, because both daylight and auroral precipitation preclude effective OH measurements.

4. Comparison of Methods

The MF radar recorded the wind speed and direction in the same height range (nominally 85 km, say ± 3 km) from which the hydroxyl emission originates (nominally 85 km) and so the same height at which one can determine the rotational temperature representing the temperature of the medium. Therefore to confirm the validity of (5), we have compared the wind speed fluctuation from MF radar measurements with that derived from the rotational temperature of hydroxyl. To this end we used measurements obtained during conditions of optimal astronomical darkness. For each day we were able to extract ~12 hours time series, as specified in Table 3. Note that by

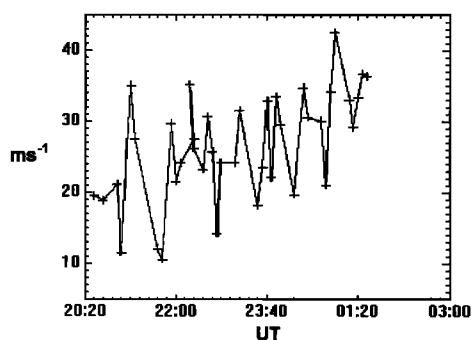


Figure 2. Time domain MF wind speed (m s^{-1}) on January 31 and February 1, 1997, for 85 km (see text for a discussion on the exactness of this height).

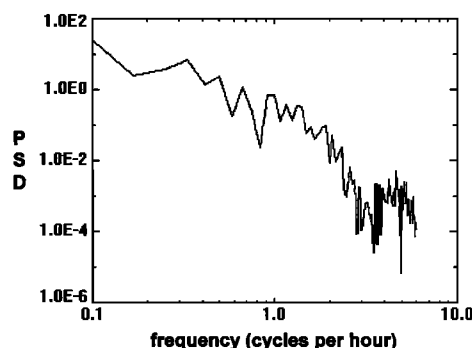


Figure 3. Spectrum of OH rotational temperature from time series in Figure 1 (January 31 and February 1, 1997). Read $1.0\text{E}2$ as 1.0×10^2 .

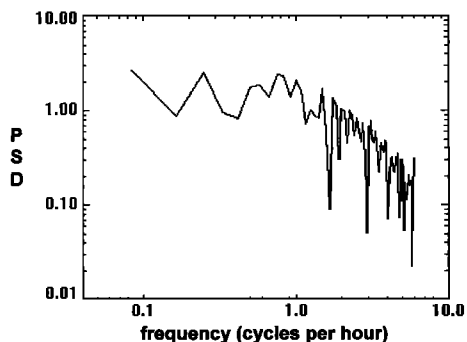


Figure 4. As for Figure 3 but for MF-derived wind (January 31 and February 1, 1997).

astronomical darkness we also imply aurora-free conditions which in turn imply both little contamination of the OH spectra and little danger of MF radar wave total reflection.

We have chosen to work with variances as indications of the perturbations because this obviates any need for consideration of phase information (equation (5) is complex). Furthermore, when considering gravity wave activity, and its relation to turbulent dissipation, it is not unusual to think in terms of variances. Let us first examine the power spectra corresponding to our temperature (OH) and wind (MF) time series. For both OH temperature and MF measurement we used fast Fourier transform with linear interpolation method. Spectra corresponding to Figures 1 and 2 are presented in Figures 3 and 4, respectively. We see a tendency to power law dependence ending in what we identify as a noise floor at high frequency. Note also that since the scales are logarithmic, the actual contribution from the noise spectral regime to the total variance is small. We excluded noise from our data using high-frequency cutoffs of 3.5 cycles/h for the OH results and 6 cycles/h for the MF results.

To identify the timescales at which (5) is valid, we have progressively filtered out increasingly higher frequencies. Thus in the time domain we applied band-pass filters to the data using a variable low-frequency cutoff and a fixed high-frequency (i.e., noise) cutoff. For each lower-frequency cutoff, we derived the corresponding horizontal velocity variance of the OH rotational temperature. In the same way we extracted the variance of the filtered MF wind for the corresponding duration. The velocity variance of both MF wind and that derived from OH rotational temperature was then plotted out

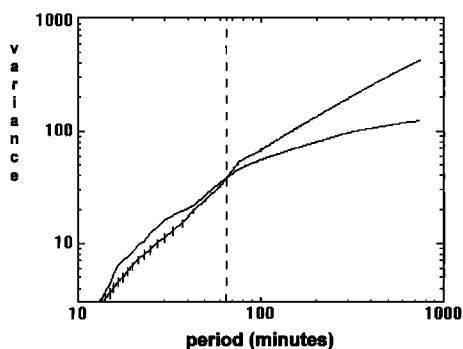


Figure 5. OH temperature perturbation and MF wind variances versus band-pass filter low-frequency cutoff (expressed in time units) plot for January 31 and February 1, 1997.

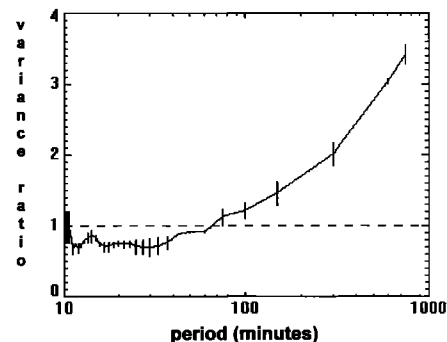


Figure 6. Ratio of (OH derived/MF derived) variance versus time for January 31 and February 1, 1997.

as a function of cutoff period, progression to shorter periods representing progressive exclusion of tidal and, eventually, longer-period GW modes. The results are shown in Figures 5–10 for 3 days of observation. Figures 11 and 12 show the average of these 3 days. In Figures 5, 7, 9, and 11 we show the variances plotted on the same axes, whereas in Figures 6, 8, 10, and 12 we show the ratio OH derived/MF derived. As we see from the figures, we can identify the regime of waves where (5) is valid, i.e., gravity waves with periods below ~ 2 hours. Although the absolute values of the variances are not identical for the two instruments, they seldom differ by more than a factor of 2 for low-frequency cutoffs corresponding to less than 120 min. As we progress to inclusion of longer periods, however, we see systematic departures, and it is these we interpret as an invalid application of *Hines* [1965]. It is somewhat subjective as to just where one should assert the simplified theory should not be used, but we have placed a conservative limit (dashed line) in Figure 5. If we are prepared to be a little less strict, one might accept periods of less than 2–3 hours as being valid for this method, from Figure 11. Our results are supported by *Manson* [1990], who noted distinctly different signatures in vertical wavelength on either side of about a 2-hour timescale.

Before leaving this discussion, we must evaluate the observational limits for each instrument. This is vital to our arguments above since we must be sure that similar parts of the dynamics spectra are being observed by each instrument. This kind of check has been described previously by, for example, *Gardner and Taylor* [1998] and implemented for real observations by *Manson* [1990] and *Manson et al.* [1998]. These studies addressed OH imagers rather than the temperature determinations used here, and also the MF radar FCA requires a

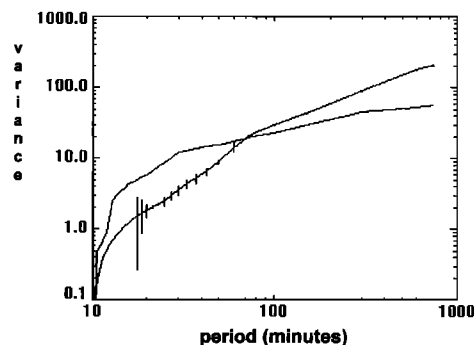


Figure 7. As for Figure 5 but for January 30 and 31, 1997.

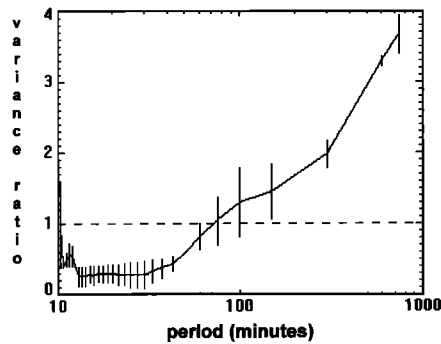


Figure 8. As for Figure 6 but for January 30 and 31, 1997.

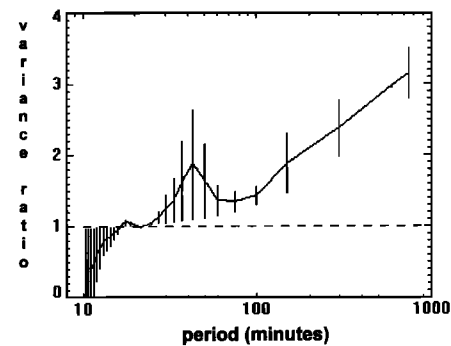


Figure 10. As for Figure 6 but for January 20 and 21, 1998.

slightly different approach (Manson *et al.* [1998] described phase/angle-of-arrival analyses). In Figure 13 we have reproduced Figure 10 of Manson *et al.* [1998], but using the format of Gardner and Taylor [1998, Figure 2]: we combine MF radar FCA and OH spectrometer observational limits along with those imposed by the atmosphere on axes of gravity wave vertical (top) and horizontal (bottom) wavelength versus period. The Brunt-Väisälä period is given by T_B , and the inertial period is given by T_i . The average diffusivity of the atmosphere is taken to be $320 \text{ m}^2 \text{ s}^{-1}$ [see Gardner and Taylor, 1998], giving the diffusive damping limit. The thickness of the OH layer multiplied by 2, λ_{OH} , gives a lower limit for vertical wavelength determination. Since we are not using an imager, the OH observation upper limit λ_{FOV} mentioned by Gardner and Taylor [1998] determined by the field of view is not applicable here. In fact, the field of view of the OH spectrometer used in this study imposes a minimum horizontal (wavelength) resolution of 9 km, although this is not indicated in Figure 13. For both instruments, there is an upper limit of observable horizontal wavelength λ_{max} , which is taken to be 100 km at a 10-min period and around 1000 km at a 100-min period. For the lower limit λ_{min} we use 20 km, this being the half-power full width of the received beam scattered from the OH height of around 85 km; Manson [1990] used 40 km for a different antenna configuration. Finally, we note the possibility that due to the nature of the electron density profile (and hence the refractive index), velocities obtained by the MF system may be weighted toward preferred heights within the 3 km range gates. We can summarize the features of Figure 13 as follows: (1) short-period cutoff at T_B (left-hand solid vertical lines) imposed by the atmosphere, (2) long-period cutoff at T_i (right-hand solid vertical lines) imposed by the atmosphere, (3) max-

imum wavelength λ_{max} (top solid lines) allowed in the atmosphere (wavelength/period cannot exceed speed of sound), (4) minimum wavelength, “diffusive damping limit” (bottom solid lines) allowed by atmospheric viscosity, (5) minimum wavelength observable by OH spectrometer λ_{OH} due to OH layer thickness (top panel horizontal/bottom panel oblique dashed line), (6) minimum wavelength observable by MF radar, λ_{min} (top panel oblique/bottom panel horizontal dotted line), (7) OH spectrometer observability: “forward” hatching, and (8) MF radar observability: “backward” hatching.

The regions of observability are depicted by the hatched areas such that areas accessible to both OH and MF radar observations are shown by cross-hatching. We see that in principle, periods of dynamics from gravity waves to the diurnal tide should be observable by both instruments, if not at all wavelengths. Recalling the sampling intervals of each instrument, and the fact that only around 5 hours of data may be compared on each night of observation, the non-cross-hatched left- and right-hand extremities may also be ignored. Thus a family of waves can exist with vertical wavelengths from 2 to 9 km, which will be included in the MF observation but not in the OH observation. Typical values of a characteristic vertical gravity wave wavelength are around 17 km [e.g., Hall and Hoppe, 1997] such that relatively few waves will exist in this “MF-only” regime, and therefore we feel that Figure 13 justifies the comparison we have made. Had we used an OH imager, a low-frequency cutoff would have been imposed at around a 300-min period. A disadvantage of the OH temperature method, however, is described in detail by Svenson and Gardner [1988], wherein spectrometer and imager methods are compared. Gravity-wave-induced temperature variations tend

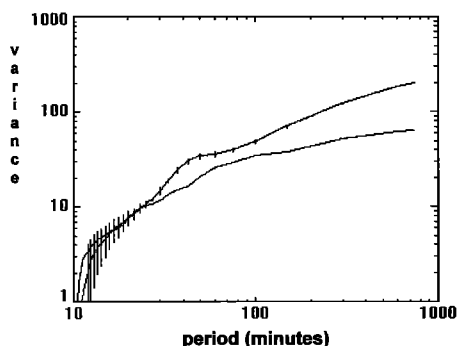


Figure 9. As for Figure 5 but for January 20 and 21, 1998.

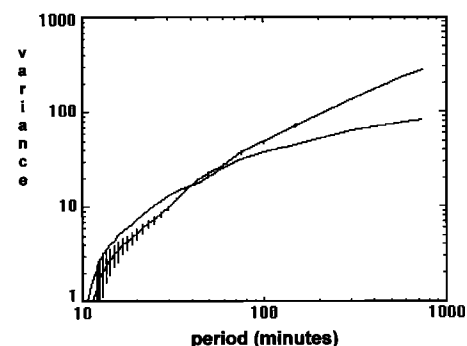


Figure 11. OH temperature perturbation and MF wind 3-day average variances versus low-frequency cutoff.

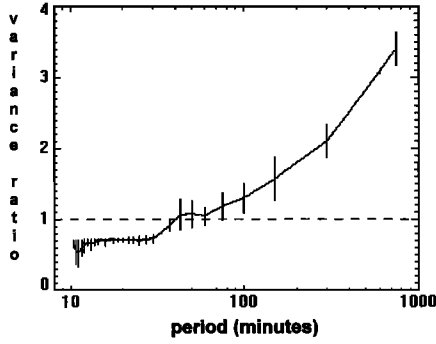


Figure 12. Ratio of the (OH derived/MF derived) variances versus low-frequency cutoff for the 3-day average.

to be underestimated by OH rotational temperature, particularly for vertical wavelengths below 15 km. *Svenson and Gardner* [1988] conclude that OH temperature measurements are most suitable for study of long-period waves, including tides,

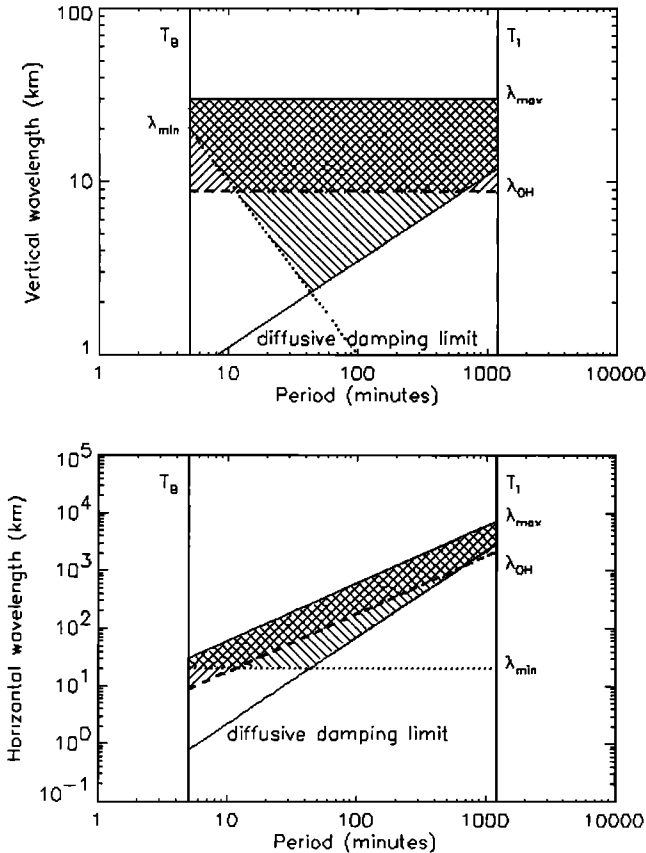


Figure 13. Observational limits for the MF radar full correlation analysis (FCA) and the OH spectrometer rotational temperature methods. (top) Gravity wave (or tide) vertical wavelength versus period. (bottom) Gravity wave (or tide) horizontal wavelength versus period. T_B is the Brunt-Väisälä period, and T_i is the inertial period. Variable λ_{OH} is the minimum vertical wavelength observable using the spectrometer due to the OH layer thickness. Variable λ_{min} is the minimum horizontal wavelength detection limit of the MF radar using FCA. Variable λ_{max} is the maximum horizontal wavelength. See Discussion and *Gardner and Taylor* [1998] for full explanation.

whereas imagers are better suited to observations of shorter periods (thus limited by λ_{FOV}). We see the effects of the underestimation of neutral temperature in our plots of variance ratio (in particular, Figures 6, 8, and 12): the OH observations clearly underestimate the perturbation detected by the MF radar.

5. Conclusions

We conclude that (5) cannot be applied directly to the derivation, from OH rotational temperature data, of the semidiurnal tide-induced contribution to the horizontal wind. Rather, it could be useful for the derivation of gravity-wave-induced horizontal wind velocity if the condition stated at the beginning of this paper is satisfied. *Myrabø and Harang* [1988] implement the *Hines* [1965] work incorrectly, i.e., for tidal-induced wind speed derivation.

The OH rotational temperature-derived horizontal wind velocity variance reported here, for gravity wave periods less than 3 hours, is in good agreement with the corresponding MF radar measurement. As we can see from Figures 6, 8, 10, and 12 the departure of OH derived from MF-derived variance is not greater than a factor of 2 for periods of less than 3 hours. This confirms that the OH-derived and MF radar measurements of horizontal wind speed variance have good agreement for shorter-period gravity waves but not tidal modes. This agreement supports the usefulness of the spectrometer for analyzing the gravity wave activity in the mesosphere provided care is taken not to work outside the requirements of the theory and particularly the assumptions inherent in its subsequent simplifications. Even so, further care must then be exercised since the true gravity wave-induced temperature variations are underestimated by OH rotational temperature measurements for short vertical wavelengths.

Appendix

We know from ideal gas law for unit mass of gas,

$$p = \rho RT, \quad (A1)$$

so differentiating, we obtain

$$dp = RTd\rho + R\rho dT. \quad (A2)$$

Dividing by the original expression,

$$\frac{dp}{p} = \frac{d\rho}{\rho} + \frac{dT}{T}. \quad (A3)$$

For a small finite change (Δp , $\Delta\rho$, ΔT) from the original state (p_0 , ρ_0 , T_0),

$$\frac{\Delta p}{p_0} - \frac{\Delta\rho}{\rho_0} = \frac{\Delta T}{T_0}. \quad (A4)$$

From (1) [*Hines*, 1960, equation (22)],

$$\frac{p - p_0}{p_0 P} = \frac{U_x}{X} \quad (A5)$$

$$\frac{p - p_0}{\rho_0 R} = \frac{U_x}{X},$$

which by rearranging gives

$$\frac{p - p_0}{p_0} = \frac{PU_x}{X}, \quad (\text{A6})$$

$$\frac{\rho - \rho_0}{\rho_0} = \frac{RU_x}{X}.$$

Subtracting and comparing with (A4), we arrive at our (4) [Hines, 1965, equation (5)]:

$$\frac{\Delta p}{p_0} - \frac{\Delta \rho}{\rho_0} = \frac{(P - R)U_x}{X} = \frac{\Delta T}{T}. \quad (\text{A7})$$

Equation (4) may be further simplified under the asymptotic condition which arises for $k_z^2 \gg \omega_a^2/C^2$ and $\omega \ll g/C$, ω_a being the acoustic period. On the basis of this the approximate dispersion relation becomes [Hines, 1960, equation (33)]

$$\omega^2 k_z^2 \approx (\omega_g^2 - \omega^2) k_x^2. \quad (\text{A8})$$

Even for low frequency this could be simplified further to [Hines, 1960; equation (34)]

$$\omega^2 k_z^2 \approx \omega_g^2 k_x^2. \quad (\text{A9})$$

If we consider all the above asymptotic conditions and approximate relations, (4) can then be further simplified as follows:

$$\begin{aligned} \frac{P}{X} &= \frac{\gamma \omega^2 \left[k_z - i \left(1 - \frac{\gamma}{2} \right) \frac{g}{C^2} \right]}{\omega k_x C^2 \left[k_z - i \left(1 - \frac{\gamma}{2} \right) \frac{g}{C^2} \right]} = \frac{\gamma \omega^2}{\omega C^2 (\omega k_z / \omega_g)} \\ &= \frac{\gamma \omega_g / C^2}{k_z} \\ &\approx 0 \text{ because } k_z \gg \frac{\omega_g}{C}, \end{aligned} \quad (\text{A10})$$

$$\begin{aligned} \frac{R}{X} &= \frac{\omega^2 k_z + i(\gamma - 1)gk_x^2 - i \frac{\gamma g \omega^2}{2C^2}}{\omega k_x C^2 \left[k_z - i \left(1 - \frac{\gamma}{2} \right) \frac{g}{C^2} \right]} = \frac{\omega^2 k_z + i(\gamma - 1)gk_x^2}{\omega k_x C^2 k_z} \\ &= \frac{\omega_g / C^2}{k_z} + i \frac{(\gamma - 1)g}{C^2 \omega_g}, \end{aligned} \quad (\text{A11})$$

$$\frac{R}{X} \approx i \frac{(\gamma - 1)g}{C^2 \sqrt{\gamma - 1} g/C},$$

$$\frac{R}{X} \approx i \sqrt{\gamma - 1} C^{-1}. \quad (\text{A12})$$

Therefore putting (A10) and (A12) into (A7),

$$\frac{\Delta T}{T} = \frac{(P - R)}{X} U_x = (0 - i \sqrt{\gamma - 1} C^{-1}) U_x, \quad (\text{A13})$$

$$\frac{\Delta T}{T} \approx -i \sqrt{\gamma - 1} C^{-1} U_x, \quad (\text{A14})$$

$$U_x = i \frac{\Delta T}{T} C (\gamma - 1)^{-1/2}. \quad (\text{A15})$$

Equation (A14) is now identical to (5) and to that of Hines [1965, equation (6)]. Equation (A15) gives, explicitly, U_x , the atmospheric gravity wave input to the horizontal wind velocity.

Variable i is the imaginary unit, the complex form being used to account for any phase information.

Notation

p	pressure (general or perturbed state).
ρ	density (general or perturbed state).
T	temperature (general or perturbed state).
p_0	pressure (initial state).
ρ_0	density (initial state).
T_0	temperature (initial state).
R	universal gas constant.
U_x	horizontal velocity perturbation.
ω	angular frequency of disturbance.
ω_g	Brunt-Väisälä frequency.
P, R, X, Z	polarization factors (see section 1).
C	speed of sound.
g	acceleration due to gravity

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